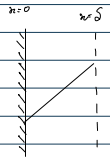
Lecture 15

Tuesday, 28 February 2023 1:31 PM

Nernst-Fickian Transport



For $M_{\nu_+}X_{\nu_-}$ where,

$z_+\nu_+ + z_-\nu_- = 0$

$c = \frac{c_+}{\nu_+} = \frac{c_-}{\nu_-}$

* Start w/ writing flux of each species:

1) $\nabla \cdot N_+ = \nabla \cdot (-z_+ u_+ F c \nu_+ - D_+ u_+ \nabla c) + \nu_+ \nabla c \cdot \bar{v}$

(Conv. of cation species; Convection)

2) $\nabla \cdot N_- = \nabla \cdot (-z_- u_- F c \nu_- - D_- u_- \nabla c) + \nu_- \nabla c \cdot \bar{v}$

* Material balance:

3) $\nu_+ \frac{\partial c}{\partial t} = -\nabla \cdot N_+ \Rightarrow \frac{\partial c}{\partial t} = -\frac{1}{\nu_+}(\nabla \cdot N_+)$

4) $\nu_- \frac{\partial c}{\partial t} = -\nabla \cdot N_- \Rightarrow \frac{\partial c}{\partial t} = -\frac{1}{\nu_-}(\nabla \cdot N_-)$

* Subtract (3-4): $0 = (z_+u_+ - z_-u_-)(F \nabla \cdot (c \nabla \phi)) + (D_+ - D_-)\nabla^2 c$

rearrange ↓

$\nabla \cdot (c \nabla \phi) = -\frac{(D_+ - D_-)}{F(z_+u_+ - z_-u_-)} \nabla^2 c$

* Plug this into: (either 3 or 4) $\frac{\partial c}{\partial t} = D \nabla^2 c - \bar{v} \cdot \nabla c$, where $D = \frac{(z_+u_+D_- - z_-u_-D_+)}{z_+D_+ - z_-D_-}$

Apply Nernst Einstein: $D_j = RTu_j$

$$\boxed{D = \frac{(z_+ - z_-) D_+ D_-}{z_+D_+ - z_-D_-}}$$

@ SS, $\frac{dc}{dt} = 0$ in 1D, $\bar{v} = 0$ (Only diff. & migration)

$$D \frac{d^2c}{dx^2} = 0$$

$c = Ax + B$; BC's: 1) @ $x = 0$, $c = 0$

$$\boxed{c = \frac{C_{bulk}\, x}{S}}$$

2) @ $x = S$, $c = C_{bulk}$

$\therefore N_+$ in BL ($\bar{v} = 0$) $= -z_+\nu_+u_+Fc\frac{d\phi}{dx} - D_+\nu_+\frac{dc}{dx}$

($\frac{dc}{dx}$: we know this; $\frac{d\phi}{dx}$: solve for this)

Since this is Cu plating, the anion does not play any role: $N_- = 0$

$\therefore 0 = -z_-\nu_-u_-Fc\frac{d\phi}{dx} - D_-\nu_+\frac{dc}{dx}$ ↓ plug in for $\frac{d\phi}{dx}$ in N_+

$$\boxed{N_+ = -\nu_+\left(\frac{z_-u_-D_+ - z_+u_+D_+}{z_-u_-}\right)\frac{dc}{dx}}$$

For $CuSO_4$, $z_+ = 2$, $z_- = -2$

$$\boxed{N_{Cu^{2+}} = -2D_{Cu^{2+}}\frac{C_{bulk,Cu^{2+}}}{S}}$$

↳ Exactly 2x the fickian diffusion

Why? + Conc. of anions has to also vary linearly in BL i.e. anions create E-field even though flux is 0, which imposes an electric field, which Cu^{2+} ions migrate within s.t. C_u has 2x flux

From transference #,

$$t_+ = \frac{z_+^2 u_+ c_+}{\sum c_j u_j z_j^2} \quad \text{(Nernst Einstein)} \qquad \text{for binary electrolyte} \Rightarrow \frac{z_+ D_+}{z_+ D_+ - z_- D_-}$$

$$t_+ = \frac{z_+ u_+}{z_+ u_+ - z_- u_-}, \quad t_- = \frac{-z_- u_-}{z_+ u_+ - z_- u_-}$$

Applying this,

$$D = \frac{z_+ u_+ D_- - z_- u_- D_+}{z_+ u_+ - z_- u_-}$$

$$\frac{N_+}{\nu_+} = -\frac{dc}{dx}\left(\frac{z_- u_- D_+ - z_+ u_+ D_-}{z_- u_-}\right)$$

$$= \frac{D(z_+ u_+ - z_- u_-)}{z_- u_-}\frac{dc}{dx}$$

$$= \frac{-D}{t_-}\frac{dc}{dx}$$

$t_- + t_+ = 1$ (fraction of total current)

$$\boxed{\frac{N_+}{\nu_+} = \frac{-D}{(1-t_+)}\frac{dc}{dx}}$$

If t_+ is small (migration becomes unimportant) !!! ★

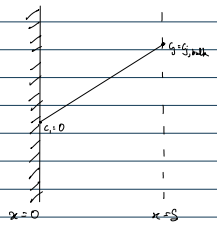
+ Often we want to add extra ions to solution (called supporting electrolyte)

↳ increase the conductivity (recall V)

↳ effect on migration → decreases electric field, means migration is less important (decrease flux to surface!)

Supporting Electrolyte (Ternary Electrolyte)

↳ Add H_2SO_4, complete dissociation to $2H^+ + SO_4^{2-}$; Cu-plating w/ $CuSO_4 + H_2SO_4$ @ $C_{Cu^{2+}} = 0$ @ surface



1 = Cu^{2+}
2 = SO_4^{2-}
3 = H^+

Dilute solution (No ion-ion, just ion-solvent)

Assume:
1) SS
2) $\bar{v} = 0$
3) Only Cu^{2+} reacts @ electrode ($N_2 = N_3 = 0$)

+ Flux of 2 and 3: (Nernst Einstein: $u \rightarrow D$)

$$\not{D}\frac{dc_j}{dx} = -\frac{c_j z_j F}{RT}\not{D}\frac{d\phi}{dx}$$

$$\frac{1}{c_j}\frac{dc_j}{dx} = -\frac{z_j F}{RT}\frac{d\phi}{dx} \qquad \left(\frac{d\ln c_j}{dx}\right)$$

Integrate (

$$\ln c_j = -\frac{z_j F}{RT}\phi + A \quad ; \quad BC: @\, x = \delta,\ \phi = 0,\ c_j = c_{j,bulk} \quad ; \quad A = \ln c_{j,bulk}$$

$$c_j = \exp\left[-\frac{z_j F \phi}{RT}\right]\exp\left[\ln c_{j,bulk}\right]$$

$$c_j = c_{j,bulk}\exp\left[-\frac{z_j F \phi}{RT}\right]$$

↳ Apply electroneutrality to get C_1:

$$z_1 C_1 = -\sum_{j=2}^{3} z_j C_j = -\sum_{j=2}^{3} z_j C_{j,bulk} \exp\left(\frac{-z_j F \phi}{RT}\right)$$

↳ Material balance for Cu:

$$\frac{\partial C_1}{\partial t} = \frac{\partial N_1}{\partial x} + R_1$$

$$\frac{\partial N_1}{\partial x} = 0 \rightarrow N_1 = \text{constant}$$

$$N_1 = -z_1 \frac{D_1 F}{RT} C_1 \frac{d\phi}{dx} - D_1 \frac{dC_1}{dx}$$

* from electroneutrality we saw $C_1 = f(\phi)$

$$\Rightarrow N_1 = D_1 \frac{d\phi}{dx}\left(-z_1 \frac{F C_1}{RT} - \frac{dC_1}{d\phi}\right) \qquad \left\{ \frac{dC_1}{dx} = \frac{dC_1}{d\phi} \cdot \frac{d\phi}{dx} \right.$$

$$\int_{x=\delta}^{x} \frac{-N_1}{D_1} dx = \int_{\phi=0}^{\phi} \left(z_1 \frac{F C_1}{RT} + \frac{dC_1}{d\phi}\right) d\phi$$

$$\frac{N_1}{D_1}(\delta - x) = \left[\frac{z_1 F}{RT}\left(\frac{-z_2}{z_1} C_{2,bulk}\left(\frac{-RT}{F}\right)\exp\left(\frac{-z_2 F\phi}{RT}\right) + \frac{-z_3}{z_1} C_{3,bulk}\left(\frac{-RT}{F z_3}\right)\exp\left(\frac{-z_3 F\phi}{RT}\right)\right) + C_1\right]_0^{\phi}$$

$\frac{-z_2}{z_1} \cdot C_{2,bulk} \cdot \exp\left(\frac{z_2 F\phi}{RT}\right) + \frac{-z_3}{z_1} C_{3,bulk} \exp\left(\frac{-z_3 F\phi}{RT}\right)$

(Use, $\frac{-z_2 C_{2,bulk} - z_3 C_{3,bulk} + C_{1,bulk}}{z_1}$)

$$\boxed{\frac{N_1}{D_1}(\delta - x) = \left(1 - \frac{z_2}{z_1}\right) C_{2,bulk} \exp\left(\frac{-z_2 F\phi}{RT}\right) + \left(1 - \frac{z_3}{z_1}\right) C_{3,bulk} \exp\left(\frac{-z_3 F\phi}{RT}\right) - C_{1,bulk} - C_{2,bulk} - C_{3,bulk}}$$

Need to only evaluate N_1 at one position (since its a constant) and satisfy $C_1 = 0$ @ $x = 0$,

$$C_1 = 0 = \frac{-1}{z_1}\left(z_2 C_{2,bulk} \exp\left(\frac{-z_2 F\phi}{RT}\right) + z_3 C_{3,bulk} \exp\left(\frac{-z_3 F\phi}{RT}\right)\right)$$

$$\phi = \frac{-RT}{F(z_2 - z_3)} \ln\left(\frac{-z_3 C_{3,bulk}}{z_2 C_{2,bulk}}\right) \quad @\ x = 0$$

↳ Plug in to integrated N_1 expression @ $x = 0$ to get N in terms of constants only.

Define a useful parameter $\Rightarrow \gamma = \frac{-z_3 C_{3,bulk}}{z_2 C_{2,bulk}}$; $\phi_0 = \frac{-RT}{F(z_2 - z_3)} \ln\gamma$

@ $x = 0$: $\frac{N_1(\delta - x)}{D_1} = \left(1 - \frac{z_2}{z_1}\right) C_{2,bulk} \exp\left(\frac{-z_2 F}{RT}\left(\frac{-RT}{F} \ln \gamma^{\frac{1}{(z_2 - z_3)}}\right)\right) + \left(1 - \frac{z_3}{z_1}\right) C_{3,bulk} \exp\left[\frac{-z_3 F}{RT}\left(\frac{-RT}{F}\right) \ln \gamma^{\frac{1}{(z_2 - z_3)}}\right] - C_{1bulk} - C_{2,bulk} - C_{3,bulk}$

$$= \left(1 - \frac{z_2}{z_1}\right) C_{2bulk} \gamma^{z_2/(z_2 - z_3)} + \left(1 - \frac{z_2}{z_1}\right) C_{3,bulk} \gamma^{z_3/(z_2 - z_3)} - C_{1b} - C_{2b} - C_{3b}$$

$$\frac{N_1 \delta}{D_1 C_{1,b}} = \frac{\frac{-z_1}{z_2}\left[\left(1 - \frac{z_2}{z_1}\right)\gamma^{z_2/(z_2 - z_3)} - \frac{z_2}{z_3}\left(1 - \frac{z_3}{z_1}\right)\gamma^{1 + \frac{z_3}{z_2 - z_3}} + \frac{z_2}{z_1}(1 - \gamma) - 1 + \frac{z_2}{z_3}\gamma\right]}{(1 - \gamma)}$$

For our electrolyte, (z_j):

$$\boxed{\frac{N_1 \delta}{D_1 C_{1,b}} = \frac{(-2 - \gamma + 3\gamma^{2/3})}{(1 - \gamma)}} \quad ; \quad \gamma = \frac{-z_3 C_{3,bulk}}{z_2 C_{2,bulk}}$$

$\gamma = 0 \rightarrow$ reduces to binary electrolyte — Have to apply L'Hopital!

↳ $\frac{N_1 \delta}{D_1 C_{1b}} = -2$

$\gamma = 1 \rightarrow$ "[illegible] electrolyte" $\Rightarrow N_1 \delta = -1$ — Main reason would be to

↳ $\frac{N_i \delta}{D_i C_{i_b}} = -2$

$\gamma = 1 \rightarrow$ "excess supporting electrolyte" $\Rightarrow \frac{N_i \delta}{D_i C_{i_b}} = -1$

↳ Expected result for diffusion only! (migration is not important)

Main reason would be to increase conductivity (improve EE of cell)
↓
You still want to keep the same kinetics by using the same electrolyte (but just reducing ohmic resistan)

So generally, when is migration important?

↳ Solve our combined current flux, eq. for $\nabla\phi$

$$\bar{i} = -F \sum_j \left(z_j^2 u_j F c_j \nabla\phi + z_j D_j \nabla c_j \right)$$

$$\nabla\phi = \frac{-1}{F \sum z_j^2 u_j c_j} \left(\frac{\bar{i}}{F} + \sum z_j D_j \nabla c_j \right)$$

Pg 120

$j \rightarrow$ species

$k \rightarrow$ all other species

$t_j \rightarrow 0$

↳ Plug into general flux expression for species k

$$N_k = \frac{t_j}{z_j F} \left(\bar{i} + F \sum_i \right)$$

* When $t_k \rightarrow 0$, migration can be neglected (for the reacting species at electrode)

Diffusion BL:

$$\delta_{diff} = \delta_{mom.} \, Sc^{-1/3} \quad ; \quad Sc = \frac{\nu}{D_j}$$

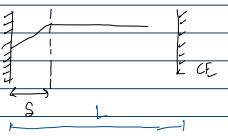
↑ navier stokes

Limiting cases for governing Eqs:

↳ Solving our transport eqs can be cumbersome and require numerical techniques to solve but we can more easily determine transport behaviour in limiting cases that are often good first approx.

1. Uniform Electrolyte composition (low current)

↳ When current is applied @ electrode, reactants are consumed, changing composition



If $\frac{i_{avg}}{i_{lim}} \ll 1$, consider uniform electrolyte

↑ current @ which $C_{surf} = 0$

* even if $\frac{i_{avg}}{i_{lim}} = 0.2$

$C_{surf} = 80\% \, C_{bulk}$

↳ If composition is uniform, how many eqs do we solve?
electroneutrality → to solve for ϕ potential in a solution

↳ Stated before $\nabla c_j = 0$, potential is defined by Laplace eq: $\nabla^2\phi = 0$

In 1D ex.: $\frac{d^2\phi}{dx^2} = 0$, subject to BC's @ electrodes

BC1 : $i_{avg} = -F \sum z_j N_j$

WE @ $x=0$

$= -F \sum z_j \left(z_j u_j F c_j \frac{d\phi}{dx} + z_i D_j \nabla c_j - c_j \vec{v} \right)$ (the ∇c_j and $\vec{v}$ terms $\to 0$)

$i_{avg} = -\kappa \frac{d\phi}{dx}$

Assume cathodic Tafel kinetics: $-\kappa \frac{d\phi}{dx} = -i_0 \exp\left(\frac{-\alpha_c F (V-\phi-U)}{RT}\right)$ → often U gets dropped (assuming that ref electrode is of same comp. as WE)

$-\kappa \frac{d\phi}{dx} = -i_0 \exp\left(\frac{\alpha_c F (V-\phi)}{RT}\right)$

BC2 (CE): @ $x=L$, $\phi = 0$

Integrating (x2) ⇒ $\phi = Ax + B$

BC ↳ $\phi = \frac{i_{avg}}{\kappa}(L-x)$, where $i_{avg} = -i_0 \exp\left[\frac{-\alpha_c F (V - \frac{i_{avg} L}{\kappa})}{RT}\right]$

We can use ϕ to tell us something about resistances that are imp. for echem performance

↳ Using ohm's law, we can define: + Charge transfer resistance (R_{ct}) → associated w/ Faradaic process @ electrode

$R_{ct} = \left.\frac{\partial(V-\phi_s)}{\partial i}\right|_{i_{avg}}$

$i_{avg} = -i_0 \exp\left(\frac{-\alpha_c F (V-\phi)}{RT}\right)$

$(V-\phi) = \frac{RT}{\alpha_c F} \ln\left(\frac{i}{i_0}\right)$

$\frac{R_{ct}}{R_\Omega}$ → charge dist.

* Slope of polarization curve, how much does current change in response to small change in overpotential

+ Ohmic resistance (solution resistance) @ electrode:
↳ associated w/ charged species moving in electrolyte

$R_\Omega = \left.\frac{\partial(\phi_0)}{\partial i}\right|_{i_{avg}} = \frac{\partial}{\partial i}\left(\frac{i_{avg}}{\kappa}(L-x)\right) = \frac{L}{\kappa}$

2. Limiting Current Density (high current)

↳ $C_{lim} = 0$ @ electrode surface

* Assume $t_{lim} \to 0$, $\vec{v} = 0$

SS $\nabla^2 c_{lim} = 0$ subject to: $c_{lim} = 0$ @ $x=0$
$c_{lim} = c_{bulk}$ @ $x = \delta$

C_{lim}

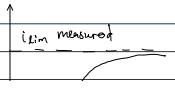
$\vec{n} \cdot \vec{N_j} = \frac{-s_j}{nF} i_{lim}$

$i_{lim} = \frac{nF}{-s_j} N_j$

$i_{lim} = \frac{nFD}{s_j} \left.\frac{dc_j}{dx}\right|_{x=0}$

$i_{lim} = \frac{nFD_{lim}}{\delta} C_{bulk}$

In practice we can compute this calc. to a measured i_{lim}



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In practice, we can compute this calc. to a measured i_{lim}

